Type MS · Define a balanced diet and explain why needs differ between people

Answer 1 of 10 · Question 2 part a · 2 marks

2024 NOV · Paper 02 · extract 1 of 2 · find part a in this mark scheme extract

2 (a)		• all components/all nutrients/all required nutrients/all food groups/protein, carbohydrates, fats, vitamins, minerals ; • in the correct quantity/correct proportion/ recommended intake;		2
(b) (i)		Protein/ /vitamins/named vitamin/minerals / named mineral/starch/fibre;		1
	(ii)	• $\frac{100}{4} = 25$; • 25×2.4; • = 60 g; **60 = 3 marks**	ECF for mp 2 and 3	3
	(iii)	• 4 slices = 470 (kJ)/1 slice = 118 (kJ); • $\frac{11\,800}{118} = 100.4$ slices; ***2 marks for 100.4 or 101or 100.43***	1 mark for 11800÷470 **or** 1 mark for 100	2
	(iv)	• break up/chop meat/ liquidise/small pieces; • place in test/boiling tube; • add alcohol/ethanol; • add water; • milky/cloudy/ milky white colouration shows presence of fat;	Apply the marking points independently Award mp3 and mp4 in any order. Accept mp3 and 4 in any order	5

Total = 13 marks

Type MS · Define a balanced diet and explain why needs differ between people

Answer 1 of 10 · Question 2 part a · 2 marks

2024 NOV · Paper 02 · extract 2 of 2 · find part a in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Define a balanced diet and explain why needs differ between people

Answer 2 of 10 · Question 2 part a · 2 marks

2019 JUN · Paper 01 (regional R) · extract 1 of 2 · find part a in this mark scheme extract

2(a)	Any two from: carbohydrates/proteins/lipids/mineral salts (or named example)/fibre;;	Ignore named examples (1 mark for each two correct)	**(2)**

Question Number	Answer	Additional guidance	Mark
2(b)	Any two from: • lack of it causes scurvy; • (lack of vitamin C causes) damage to membranes/bleeding gums; • makes connective tissue/maintains/strengthens cell linings/gums/sticks/joins cells/tissues together;		**(2)**

Question Number	Answer	Additional guidance	Mark
2(c)(i)	Any five from: • Add fresh juice/long-life juice (to a test tube); • add DCPIP drop by drop/slowly; • until blue colour disappears/solution turns colourless; • count number of drops/measure volume of DCPIP added; • repeat (to obtain mean); • compare volumes of DCPIP added;		**(5)**

Question Number	Answer	Additional guidance	Mark
2(c)(ii)	Any two from: • volume of juice; • temperature; • concentration DCPIP;	Allow amount for mp1. Ignore concentration	**(2)**

Total 11 marks

Type MS · Define a balanced diet and explain why needs differ between people

Answer 2 of 10 · Question 2 part a · 2 marks

2019 JUN · Paper 01 (regional R) · extract 2 of 2 · find part a in this mark scheme extract

Paper 1: Subject Human Biology

Question Number	Answer	Additional guidance	Mark

Type MS · Define a balanced diet and explain why needs differ between people

Answer 3 of 10 · Question 4 part b · 2 marks

2023 JUN · Paper 01 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

Question number				
4	(a)	• correct plots;;; • key/labels;		4
	(b)	• correct components/nutrients/list of components min of five; • in correct amounts/proportions;		2
	(c)	• 75; • 15 • = 5:1;	ECF	2
	(d)	• contain protein • needed for growth/repair/,muscles/tissues; • area A girl has low(er) levels/area B girl has high(er) levels; • less/poor growth/area B girl good growth; • kwashiorkor;		5

Total 13 marks

Type MS · Define a balanced diet and explain why needs differ between people

Answer 3 of 10 · Question 4 part b · 2 marks

2023 JUN · Paper 01 (regional R) · extract 2 of 2 · find part b in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Define a balanced diet and explain why needs differ between people

Answer 4 of 10 · Question 3 part a.i · 2 marks

2024 JUN · Paper 01 (regional R) · find part a.i in this mark scheme extract

3 (a) (i)	contains all nutrients needed/ correct balance of nutrients needed/ correct proportion of nutrients; to maintain health;	If groups of nutrients given all must be named	(2)
(ii)	males need more energy than females; 3000 kJ; greater muscle mass; males are likely to be more active/ higher metabolic rate:	Allow reverse for female	(3)
(iii)	iron needed to form haemoglobin; to replace haemoglobin lost during menstruation/red blood cells are lost during menstruation;		(2)
(iv)	vitamin A;		(1)
(v)	In 250g red meat - 2.5 x 2.5 = 6.25; In 200g broccoli - 2 x 0.7 = 1.4; 12 - 7.65; 4.35 (mg);	Full marks for correct final answer ecf for mp3 and mp4	(4)
(b) (i)	bleeding gums/anaemia/bruising/poor wound healing;	Allow other valid symptoms	(1)
(ii)	more fruit/vegetables eaten in diet/increased intake of vitamin C/ vitamin c supplement;		(1)

Total for Question 3 = 14 marks

Type MS · Define a balanced diet and explain why needs differ between people

Answer 5 of 10 · Question 2 part b.i · 1 mark

2024 NOV · Paper 02 · extract 1 of 2 · find part b.i in this mark scheme extract

2 (a)		• all components/all nutrients/all required nutrients/all food groups/protein, carbohydrates, fats, vitamins, minerals ; • in the correct quantity/correct proportion/ recommended intake;		2
(b) (i)		Protein/ /vitamins/named vitamin/minerals / named mineral/starch/fibre;		1
	(ii)	• $\frac{100}{4} = 25$; • 25×2.4; • = 60 g; **60 = 3 marks**	ECF for mp 2 and 3	3
	(iii)	• 4 slices = 470 (kJ)/1 slice = 118 (kJ); • $\frac{11\,800}{118} = 100.4$ slices; ***2 marks for 100.4 or 101or 100.43***	1 mark for 11800÷470 **or** 1 mark for 100	2
	(iv)	• break up/chop meat/ liquidise/small pieces; • place in test/boiling tube; • add alcohol/ethanol; • add water; • milky/cloudy/ milky white colouration shows presence of fat;	Apply the marking points independently Award mp3 and mp4 in any order. Accept mp3 and 4 in any order	5

Total = 13 marks

Type MS · Define a balanced diet and explain why needs differ between people

Answer 5 of 10 · Question 2 part b.i · 1 mark

2024 NOV · Paper 02 · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Define a balanced diet and explain why needs differ between people

Answer 6 of 10 · Question 2 part b · 1 mark

2020 JAN · Paper 01 · find part b in this mark scheme extract

number				
2 (a)	(i)	**The only correct answer is A** *B is incorrect because it is the cervix* *C is incorrect because it is the oviduct* *D is incorrect because it is the uterus wall*		1
	(ii)	**The only correct answer is D** *A is incorrect because it is the vagina* *B is incorrect because it is the cervix* *C is incorrect because it is the oviduct*		1
	(iii)	• oestrogen (1) • lining (1) • ovulation (1)		3
(b)		needed for the growth of bones (1)		1
(c)		ref antibodies (passed to baby)/contains the correct balance of nutrients/reduced risk of diabetes/allergies/obesity later in life/reference to bonding between mother and baby (1)		1
(d)	(i)	more breastfed babies in 2010 compared with 1940 (1)	Allow reverse argument A 78% vs 42% difference of 36%	1
	(ii)	• life expectancy shows a constant increase (1) • despite number of babies breastfed increasing and decreasing (1)		2

Total for Question 2 = 10 marks

Type MS · Define a balanced diet and explain why needs differ between people

Answer 7 of 10 · Question 5 part a.ii · 4 marks

2020 NOV · Paper 01 (regional R) · find part a.ii in this mark scheme extract

5 (a)	(i)	**C** green vegetables and red meat		1
	(ii)	• (iron) needed for production of haemoglobin(1) • (haemoglobin/red blood cell) binds to/transports oxygen(1) • oxygen needed for (aerobic) respiration/energy(1) • for growth of fetus(1)		4
(b)	(i)	Any four from: • take temperature of water (at the start) (1) • reference to measuring mass of food(1) • place food on holder(1) • burn food(1) • reference to time (for burning food)(1) • take temperature of water after burning/once food is completely burnt/measure change in temperature(1)	Allow amount for mass Allow food is heated/set alight	4
	(ii)	Any three from: • add a lid(1) • reference to stirring/even heat distribution(1) • insulate the copper can/move food closer (to the can/water) (1) • reference to preventing heat loss(1)	Allow cover the beaker	3

Total for question 5 = 12 marks

Type MS · Define a balanced diet and explain why needs differ between people

Answer 8 of 10 · Question 3 part b.ii · 1 mark

2022 JUN · Paper 01 · extract 1 of 2 · find part b.ii in this mark scheme extract

3 (a) (i)	Any five from the following: • place water in the test tube; • take the (starting) temperature of the water; • attach the test tube to a clamp stand; • measure the mass of a piece of food/weigh food; • place food on the mounted needle (under the test tube); • burn the food; (using a Bunsen burner) • take the (final) temperature of the water;		5
(ii)	Any two from the following: • volume/mass of water; • distance of the food from the test tube; • mass of food;	ignore amount ignore amount/size	2
(iii)	Any two from the following: • hold the burning food at arm's length; • reference to sharpness of needle; • goggles; • reference to food allergies/don't taste food;	any ref to being burnt/hot water	2
(b) (i)	Any two from the following: • males need more energy than females; • the amount of energy required decreases with age; • the amount of energy needed by males decreases more (overall) than females;		2
(ii)	**D** - Males use more muscle mass than females;		1
(iii)	• (2629 – 2103) = 526; • 526 ÷ 2103; • (x 100) = 25 (%);	if divide by 2629 and give 20% then two marks Total 15 marks	3

Type MS · Define a balanced diet and explain why needs differ between people

Answer 8 of 10 · Question 3 part b.ii · 1 mark

2022 JUN · Paper 01 · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Define a balanced diet and explain why needs differ between people

Answer 9 of 10 · Question 3 part a.ii · 3 marks

2024 JUN · Paper 01 (regional R) · find part a.ii in this mark scheme extract

3 (a) (i)	contains all nutrients needed/ correct balance of nutrients needed/ correct proportion of nutrients; to maintain health;	If groups of nutrients given all must be named	(2)
(ii)	males need more energy than females; 3000 kJ; greater muscle mass; males are likely to be more active/ higher metabolic rate:	Allow reverse for female	(3)
(iii)	iron needed to form haemoglobin; to replace haemoglobin lost during menstruation/red blood cells are lost during menstruation;		(2)
(iv)	vitamin A;		(1)
(v)	In 250g red meat - 2.5 x 2.5 = 6.25; In 200g broccoli - 2 x 0.7 = 1.4; 12 - 7.65; 4.35 (mg);	Full marks for correct final answer ecf for mp3 and mp4	(4)
(b) (i)	bleeding gums/anaemia/bruising/poor wound healing;	Allow other valid symptoms	(1)
(ii)	more fruit/vegetables eaten in diet/increased intake of vitamin C/ vitamin c supplement;		(1)

Total for Question 3 = 14 marks

Type MS · Define a balanced diet and explain why needs differ between people

Answer 10 of 10 · Question 3 part a.iii · 2 marks

2024 JUN · Paper 01 (regional R) · find part a.iii in this mark scheme extract

3 (a) (i)	contains all nutrients needed/ correct balance of nutrients needed/ correct proportion of nutrients; to maintain health;	If groups of nutrients given all must be named	(2)
(ii)	males need more energy than females; 3000 kJ; greater muscle mass; males are likely to be more active/ higher metabolic rate:	Allow reverse for female	(3)
(iii)	iron needed to form haemoglobin; to replace haemoglobin lost during menstruation/red blood cells are lost during menstruation;		(2)
(iv)	vitamin A;		(1)
(v)	In 250g red meat - 2.5 x 2.5 = 6.25; In 200g broccoli - 2 x 0.7 = 1.4; 12 - 7.65; 4.35 (mg);	Full marks for correct final answer ecf for mp3 and mp4	(4)
(b) (i)	bleeding gums/anaemia/bruising/poor wound healing;	Allow other valid symptoms	(1)
(ii)	more fruit/vegetables eaten in diet/increased intake of vitamin C/ vitamin c supplement;		(1)

Total for Question 3 = 14 marks